

Apoorva Nema

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EDUCATION

Vellore Institute of Technology, B.Tech in Computer Science and Engineering Aug 2022 – 2026
CGPA: 8.34/10

Kendriya Vidyalaya DRDO, 12th (CBSE), Bangalore 2020 - 2022
Percentage: 88.8%

Kendriya Vidyalaya DRDO, 10th (CBSE), Bangalore 2010 - 2020
Percentage: 93.2%

WORK EXPERIENCE

Centre for AI & Robotics (DRDO), Research Intern Jun 2024 – Jul 2024
– Developed an end-to-end deepfake speech detection ML pipeline for 100+ synthetic and real audio samples, performing preprocessing, feature extraction, generating Vision Transformer embeddings on spectrograms, applying clustering and distance metrics for separation, and benchmarking models for high-accuracy real vs AI audio classification.

Inferigence Quotient Pvt Ltd, AI Intern Jun 2023 – Jul 2023
– Built and optimized an NLP workflow for speech transcript sentiment classification using preprocessing, vectorization, feature engineering, and ML modeling.

PROJECTS

Dark Detect Chrome Extension [GitHub]
– Built a Chrome extension that detects and recognizes deceptive UI patterns using Naive Bayes on ML extracted DOM features, achieving 85% accuracy on benchmark datasets, with the classifier deployed separately via Streamlit.

Domain Mentor [GitHub]
– Engineered semantic similarity scoring and automated matching pipeline using WordNet to recommend research domains.

Multispectral Camouflage Detection System [Colab Link]
– Developed a dual-branch ML system combining YOLO detection and ResNet classification for low-contrast camouflaged objects, integrated image preprocessing and augmentation pipeline.

SKILLS

Programming & CS: Python, Java, SQL, Data Structures & Algorithms, Object Oriented Programming, Operating Systems, Database Management Systems, Networks, Compiler Design, Linux, Cybersecurity, Software Engineering

AI/ML: Supervised and Unsupervised Machine Learning, NLP, Deep Learning, Exploratory Data Analysis, Computer Vision, Predictive Analytics, Data Visualization, Generative AI

Frameworks: PyTorch, TensorFlow, Keras, Flask, NumPy, Pandas

Tools: AWS, Git, Linux, Google Colab

Languages: English (IELTS 7.5), Spanish (A1)

CERTIFICATIONS

Machine Learning A–Z (Udemy)

LEADERSHIP & ACTIVITIES

Senior Core Member in roboVITics : Mentored 20 members, reviewed AI projects and led automation focused workshops.

Youth Parliament Delegate : National-level debating

ACHIEVEMENTS

- Top 5% in Adobe Gensolve Hackathon 2024
- Solv-a-thon Finalist (Top 10 across 4 campuses)
- DRDO Meritorious Award for Academic Excellence